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LOW LOSS TRANSVERSELY-EXCITED FILM BULK ACOUSTIC RESONATORS AND FILTERS

Abstract

An acoustic resonator device is provided that includes a piezoelectric layer; and a conductor pattern on a surface of the piezoelectric layer. The conductor pattern includes an interdigital transducer that includes a first busbar, a second busbar, and interleaved fingers that each extend from one of the first and second busbars. The interleaved fingers includes a first finger and a last finger at opposing ends of the IDT. The conductor pattern includes a first reflector element proximate to the first finger; and a second reflector element proximate to the last finger. A distance between the first reflector element and the first finger is greater than a pitch p of the IDT, the pitch p being a center-to-center spacing between adjacent interleaved fingers. Moreover, a distance between the second reflector element and the last finger is greater than the pitch p of the IDT.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/855,711, filed Jun. 30, 2022, which claims priority to U.S. Provisional Patent Application No. 63/244,565, filed Sep. 15, 2021, entitled XBAR FILTER END REFLECTORS. U.S. patent application Ser. No. 17/855,711 is also a continuation-in-part of U.S. patent application Ser. No. 17/555,349, filed Dec. 17, 2021, entitled LOW LOSS TRANSVERSELY-EXCITED FILM BULK ACOUSTIC RESONATORS AND FILTERS. The contents of each of these applications are hereby incorporated by reference in their entireties. [0002] U.S. patent application Ser. No. 17/555,349 claims priority to U.S. Provisional Patent Application No. 63/196,645, filed Jun. 3, 2021, entitled XBAR N79 FILTER GRATING ELEMENTS. U.S. patent application Ser. No. 17/555,349 is also a continuation-in-part of U.S. patent application Ser. No. 17/229,767, filed Apr. 13, 2021, now issued as U.S. Pat. No. 12,040,779, which claims priority to the following provisional patent applications: U.S. Application No. 63/012,849, filed Apr. 20, 2020, entitled SMALL HIGH Q XBAR RESONATORS; U.S. Application No. 63/066,520, filed Aug. 17, 2020, entitled SMALL REFLECTORS TO IMPROVE XBAR LOSS; and U.S. Application No. 63/074,991, filed Sep. 4, 2020, entitled SMALL REFLECTORS TO IMPROVE PERFORMANCE OF TRANSVERSELY-EXCITED FILM BULK ACOUSTIC RESONATORS AT A SPECIFIED FREQUENCY. The contents of each of these applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0003] This disclosure relates to radio frequency filters using acoustic wave resonators, and specifically to filters for use in communications equipment.

BACKGROUND

[0004] A radio frequency (RF) filter is a two-port device configured to pass some frequencies and to stop other frequencies, where “pass” means transmit with relatively low signal loss and “stop” means block or substantially attenuate. The range of frequencies passed by a filter is referred to as the “pass-band” of the filter. The range of frequencies stopped by such a filter is referred to as the “stop-band” of the filter. A typical RF filter has at least one pass-band and at least one stop-band. Specific requirements on a pass-band or stop-band depend on the application. For example, a “pass-band” may be defined as a frequency range where the insertion loss of a filter is better than a defined value such as 1 dB, 2 dB, or 3 dB. A “stop-band” may be defined as a frequency range where the rejection of a filter is greater than a defined value such as 20 dB, 30 dB, 40 dB, or greater depending on application.

[0005] RF filters are used in communications systems where information is transmitted over

wireless links. For example, RF filters may be found in the RF front-ends of cellular base stations, mobile telephone and computing devices, satellite transceivers and ground stations, IoT (Internet of Things) devices, laptop computers and tablets, fixed point radio links, and other communications systems. RF filters are also used in radar and electronic and information warfare systems.

[0006] RF filters typically require many design trade-offs to achieve, for each specific application, the best compromise between performance parameters such as insertion loss, rejection, isolation, power handling, linearity, size and cost. Specific design and manufacturing methods and enhancements can benefit simultaneously one or several of these requirements.

[0007] Performance enhancements to the RF filters in a wireless system can have broad impact to system performance. Improvements in RF filters can be leveraged to provide system performance improvements such as larger cell size, longer battery life, higher data rates, greater network capacity, lower cost, enhanced security, higher reliability, etc. These improvements can be realized at many levels of the wireless system both separately and in combination, for example at the RF module, RF transceiver, mobile or fixed sub-system, or network levels.

[0008] High performance RF filters for present communication systems commonly incorporate acoustic wave resonators including surface acoustic wave (SAW) resonators, bulk acoustic wave (BAW) resonators, film bulk acoustic wave resonators (FBAR), and other types of acoustic resonators. However, these existing technologies are not well-suited for use at the higher frequencies and bandwidths proposed for future communications networks.

[0009] The desire for wider communication channel bandwidths will inevitably lead to the use of higher frequency communications bands. Radio access technology for mobile telephone networks has been standardized by the 3GPP (3.sup.rd Generation Partnership Project). Radio access technology for 5.sup.th generation (5G) mobile networks is defined in the 5G NR (new radio) standard. The 5G NR standard defines several new communications bands. Two of these new communications bands are n77, which uses the frequency range from 3300 MHz to 4200 MHz, and n79, which uses the frequency range from 4400 MHz to 5000 MHz. Both band n77 and band n79 use time-division duplexing (TDD), such that a communications device operating in band n77 and/or band n79 use the same frequencies for both uplink and downlink transmissions. Bandpass filters for bands n77 and n79 must be capable of handling the transmit power of the communications device. WiFi bands at 5 GHz and 6 GHz also require high frequency and wide bandwidth. The 5G NR standard also defines millimeter wave communication bands with frequencies between 24.25 GHz and 40 GHz.

[0010] The Transversely-Excited Film Bulk Acoustic Resonator (XBAR) is an acoustic resonator structure for use in microwave filters. The XBAR is described in patent U.S. Pat. No. 10,491,291, titled TRANSVERSELY EXCITED FILM BULK ACOUSTIC RESONATOR. An XBAR resonator comprises an interdigital transducer (IDT) formed on a thin floating layer, or diaphragm, of a single-crystal piezoelectric material. The IDT includes a first set of parallel fingers, extending from a first busbar and a second set of parallel fingers extending from a second busbar. The first and second sets of parallel fingers are interleaved. A microwave signal applied to the IDT excites a shear primary acoustic wave in the piezoelectric diaphragm. XBAR resonators provide very high electromechanical coupling and high frequency capability. XBAR resonators may be used in a variety of RF filters including band-reject filters, band-pass filters, duplexers, and multiplexers. XBARs are well suited for use in filters for communications bands with frequencies above 3 GHz.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 includes a schematic plan view, two schematic cross-sectional views, and a detail view of a transversely-excited film bulk acoustic resonator (XBAR).

[0012] FIG. **2** is a schematic block diagram of a band-pass filter using acoustic resonators.

[0013] FIG. **3** is a graph of relationships between the Q-factor of an XBAR and the number of fingers in the interdigital transducer (IDT) of the XBAR.

[0014] FIG. **4** is a schematic plan view of an IDT with reflector elements.

[0015] FIG. **5** is a schematic plan view of another IDT with reflector elements.

[0016] FIG. **6** is a graph comparing the normalized Q-factor of an XBAR with and without reflector elements at the resonance frequency.

[0017] FIG. **7** is a graph comparing the normalized Q-factor of an XBAR with and without reflector elements at the anti-resonance frequency.

[0018] FIG. **8** is a graph showing relative Q-factor as a function of reflector element pitch and mark for a representative XBAR at a frequency of 5150 MHz.

[0019] FIG. **9** is a graph showing relative Q-factor as a function of frequency and reflector element mark for an XBAR with two reflector elements at each end.

[0020] FIG. **10** is a graph showing relative Q-factor as a function of frequency and reflector element mark for an XBAR with one reflector element at each end.

[0021] FIG. **11** is a graph showing relative Q-factor as a function of frequency and reflector element mark for an XBAR with five reflector elements at each end.

[0022] FIG. **12** is a graph comparing the performance of two band-pass filters using XBARs with and without reflector elements.

[0023] FIG. **13** is a schematic plan view of another IDT with reflector elements.

[0024] FIG. **14A** and FIG. **14B** are graphs comparing the performance of band-pass filters using XBARs with and without reflector elements.

[0025] FIG. **15** is a flow chart of a method for fabricating an XBAR or a filter using XBARs.

[0026] Throughout this description, elements appearing in figures are assigned three-digit or four-digit reference designators, where the two least significant digits are specific to the element and the one or two most significant digit is the figure number where the element is first introduced. An element that is not described in conjunction with a figure may be presumed to have the same characteristics and function as a previously-described element having the same reference designator.

DETAILED DESCRIPTION

Description of Apparatus

[0027] FIG. **1** shows a simplified schematic top view and orthogonal cross-sectional views of an XBAR **100**. XBAR resonators such as the resonator **100** may be used in a variety of RF filters including band-reject filters, band-pass filters, duplexers, and multiplexers.

[0028] The XBAR **100** is made up of a thin film conductor pattern formed on a surface of a piezoelectric plate **110** having parallel front and back surfaces **112**, **114**, respectively. The piezoelectric plate is a thin single-crystal layer of a piezoelectric material such as lithium niobate, lithium tantalate, lanthanum gallium silicate, gallium nitride, or aluminum nitride. The piezoelectric plate is cut such that the orientation of the X, Y, and Z crystalline axes with respect to the front and back surfaces is known and consistent. The piezoelectric plate may be Z-cut, which is to say the Z axis is normal to the front and back surfaces **112**, **114**. The piezoelectric plate may be rotated Z-cut or rotated YX-cut. XBARs may be fabricated on piezoelectric plates with other crystallographic orientations.

[0029] The back surface **114** of the piezoelectric plate **110** is attached to a surface of a substrate **120** except for a portion of the piezoelectric plate **110** that forms a diaphragm **115** spanning a cavity **140** formed in the substrate. The portion of the piezoelectric plate that spans the cavity is referred to herein as the “diaphragm” **115** due to its physical resemblance to the diaphragm of a microphone. As shown in FIG. **1**, the diaphragm **115** is contiguous with the rest of the piezoelectric plate **110** around all of a perimeter **145** of the cavity **140**. In this context, “contiguous” means “continuously connected without any intervening item”. In other configurations, the diaphragm **115** may be

contiguous with the piezoelectric plate around at least 50% of the perimeter **145** of the cavity **140**. [0030] The substrate **120** provides mechanical support to the piezoelectric plate **110**. The substrate **120** may be, for example, silicon, sapphire, quartz, or some other material or combination of materials. The back surface **114** of the piezoelectric plate **110** may be bonded to the substrate **120** using a wafer bonding process. Alternatively, the piezoelectric plate **110** may be grown on the substrate **120** or attached to the substrate in some other manner. The piezoelectric plate **110** may be attached directly to the substrate or may be attached to the substrate **120** via one or more intermediate material layers (not shown in FIG. 1).

[0031] “Cavity” has its conventional meaning of “an empty space within a solid body.” The cavity **140** may be a hole completely through the substrate **120** (as shown in Section A-A and Section B-B) or a recess in the substrate **120** under the diaphragm **115**. The cavity **140** may be formed, for example, by selective etching of the substrate **120** before or after the piezoelectric plate **110** and the substrate **120** are attached.

[0032] The conductor pattern of the XBAR **100** includes an interdigital transducer (IDT) **130**. The IDT **130** includes a first plurality of parallel fingers, such as finger **136**, extending from a first busbar **132** and a second plurality of fingers extending from a second busbar **134**. The term “busbar” means a conductor from which the fingers of an IDT extend. The first and second pluralities of parallel fingers are interleaved. The interleaved fingers overlap for a distance AP, commonly referred to as the “aperture” of the IDT. The center-to-center distance L between the outermost fingers of the IDT **130** is the “length” of the IDT.

[0033] The first and second busbars **132**, **134** serve as the terminals of the XBAR **100**. A radio frequency or microwave signal applied between the two busbars **132**, **134** of the IDT **130** excites a primary acoustic mode within the piezoelectric plate **110**. The primary acoustic mode is a bulk shear mode where acoustic energy propagates along a direction substantially orthogonal to the surface of the piezoelectric plate **110**, which is also normal, or transverse, to the direction of the electric field created by the IDT fingers. Thus, the XBAR is considered a transversely-excited film bulk wave resonator.

[0034] The IDT **130** is positioned on the piezoelectric plate **110** such that at least the fingers of the IDT **130** are disposed on the diaphragm **115** that spans, or is suspended over, the cavity **140**. As shown in FIG. 1, the cavity **140** has a rectangular shape with an extent greater than the aperture AP and length L of the IDT **130**. A cavity of an XBAR may have a different shape, such as a regular or irregular polygon. The cavity of an XBAR may have more or fewer than four sides, which may be straight or curved.

[0035] For ease of presentation in FIG. 1, the geometric pitch and width of the IDT fingers are greatly exaggerated with respect to the length (dimension L) and aperture (dimension AP) of the XBAR. A typical XBAR has more than ten parallel fingers in the IDT **130**. An XBAR may have hundreds, possibly thousands, of parallel fingers in the IDT **130**. Similarly, the thicknesses of the IDT fingers and the piezoelectric plate in the cross-sectional views are greatly exaggerated.

[0036] Referring now to the detailed schematic cross-sectional view, a front-side dielectric layer **150** may optionally be formed on the front side of the piezoelectric plate **110**. The “front side” of the XBAR is, by definition, the surface facing away from the substrate. The front-side dielectric layer **150** may be formed only between the IDT fingers (e.g. IDT finger **138b**) or may be deposited as a blanket layer such that the dielectric layer is formed both between and over the IDT fingers (e.g. IDT finger **138a**). The front-side dielectric layer **150** may be a non-piezoelectric dielectric material, such as silicon dioxide, alumina, or silicon nitride. A thickness of the front side dielectric layer **150** is typically less than about one-third of the thickness t_p of the piezoelectric plate **110**. The front-side dielectric layer **150** may be formed of multiple layers of two or more materials. In some applications, a back-side dielectric layer (not shown) may be formed on the back side of the piezoelectric plate **110**.

[0037] The IDT fingers **138a**, **138b** may be one or more layers of aluminum, an aluminum alloy,

copper, a copper alloy, beryllium, gold, tungsten, molybdenum, chromium, titanium or some other conductive material. The IDT fingers are considered to be “substantially aluminum” if they are formed from aluminum or an alloy comprising at least 50% aluminum. The IDT fingers are considered to be “substantially copper” if they are formed from copper or an alloy comprising at least 50% copper. Thin (relative to the total thickness of the conductors) layers of metals such as chromium or titanium may be formed under and/or over and/or as layers within the fingers to improve adhesion between the fingers and the piezoelectric plate **110** and/or to passivate or encapsulate the fingers and/or to improve power handling. The busbars (**132**, **134** in FIG. **1**) of the IDT may be made of the same or different materials as the fingers.

[0038] Dimension p is the center-to-center spacing or “pitch” of the IDT fingers, which may be referred to as the pitch of the IDT and/or the pitch of the XBAR. Dimension m is the width or “mark” of the IDT fingers. In some embodiments the pitch and/or the mark of the IDT fingers may vary slightly along the length of the IDT. In such cases, the dimensions p and m are the average of the pitch and mark, respectively. The geometry of the IDT of an XBAR differs substantially from the IDTs used in surface acoustic wave (SAW) resonators. In a SAW resonator, the pitch of the IDT is one-half of the acoustic wavelength at the resonance frequency. Additionally, the mark-to-pitch ratio of a SAW resonator IDT is typically close to 0.5 (i.e. the mark or finger width is about one-fourth of the acoustic wavelength at resonance). In an XBAR, the pitch p of the IDT may be 2 to 20 times the width w of the fingers. The pitch p is typically 3.3 to 5 times the width w of the fingers. In addition, the pitch p of the IDT may be 2 to 20 times the thickness of the piezoelectric plate **210**. The pitch p of the IDT is typically 5 to 12.5 times the thickness of the piezoelectric plate **210**. The width m of the IDT fingers in an XBAR is not constrained to be near one-fourth of the acoustic wavelength at resonance. For example, the width of XBAR IDT fingers may be 500 nm or greater, such that the IDT can be readily fabricated using optical lithography. The thickness of the IDT fingers may be from 100 nm to about equal to the width m . The thickness of the busbars (**132**, **134**) of the IDT may be the same as, or greater than, the thickness t_m of the IDT fingers.

[0039] FIG. **2** is a schematic circuit diagram and layout for a high frequency band-pass filter **200** using XBARs. The filter **200** has a conventional ladder filter architecture including three series resonators **210A**, **210B**, **210C** and two shunt resonators **220A**, **220B**. The three series resonators **210A**, **210B**, and **210C** are connected in series between a first port and a second port (hence the term “series resonator”). In FIG. **2**, the first and second ports are labeled “In” and “Out”, respectively. However, the filter **200** is bidirectional and either port may serve as the input or output of the filter. The two shunt resonators **220A**, **220B** are connected from nodes between the series resonators to ground. A filter may contain additional reactive components, such as capacitors and/or inductors, not shown in FIG. **2**. All the shunt resonators and series resonators are XBARs. The inclusion of three series and two shunt resonators is exemplary. A filter may have more or fewer than five total resonators, more or fewer than three series resonators, and more or fewer than two shunt resonators. Typically, all of the series resonators are connected in series between an input and an output of the filter. All of the shunt resonators are typically connected between ground and the input, the output, or a node between two series resonators.

[0040] In the exemplary filter **200**, the three series resonators **210A**, **B**, **C** and the two shunt resonators **220A**, **B** of the filter **200** are formed on a single plate **230** of piezoelectric material bonded to a silicon substrate (not visible). In some filters, the series resonators and shunt resonators may be formed on different plates of piezoelectric material. Each resonator includes a respective IDT (not shown), with at least the fingers of the IDT disposed over a cavity in the substrate. In this and similar contexts, the term “respective” means “relating things each to each”, which is to say with a one-to-one correspondence. In FIG. **2**, the cavities are illustrated schematically as the dashed rectangles (such as the rectangle **235**). In this example, each IDT is disposed over a respective cavity. In other filters, the IDTs of two or more resonators may be disposed over a single cavity.

[0041] Each of the resonators **210A**, **210B**, **210C**, **220A**, **220B** in the filter **200** has resonance

where the admittance of the resonator is very high and an anti-resonance where the admittance of the resonator is very low. The resonance and anti-resonance occur at a resonance frequency and an anti-resonance frequency, respectively, which may be the same or different for the various resonators in the filter **200**. In over-simplified terms, each resonator can be considered a short-circuit at its resonance frequency and an open circuit at its anti-resonance frequency. The input-output transfer function will be near zero at the resonance frequencies of the shunt resonators and at the anti-resonance frequencies of the series resonators. In a typical filter, the resonance frequencies of the shunt resonators are positioned below the lower edge of the filter's passband and the anti-resonance frequencies of the series resonators are positioned above the upper edge of the passband. In some filters, a front-side dielectric layer (also called a “frequency setting layer”), represented by the dot-dash rectangle **270**, may be formed on the shunt resonators to set the resonance frequencies of the shunt resonators lower relative to the resonance frequencies of the series resonators.

[0042] The Q-factor of an acoustic resonator is commonly defined as the peak energy stored during a cycle of the applied RF signal divided by the total energy dissipated or lost during the cycle. The Q-factor of an XBAR is a complex function of numerous parameters including the length, or number of fingers, in the IDT of the XBAR.

[0043] Possible loss mechanisms in an acoustic resonator include resistive losses in the IDT and other conductors; viscous or acoustic losses in the piezoelectric plate, IDT fingers, and other materials; and leakage of acoustic energy out of the resonator structure. The peak energy stored in a resonator is proportional the capacitance of the resonator. In an XBAR resonator, the capacitance is proportional to the number of IDT fingers. Resistive losses and viscose losses are also proportional to the number of IDT fingers. Acoustic energy that leaks from the resonator in the transverse direction (i.e. the direction parallel to the IDT fingers) is proportional to the length of the resonator and thus also proportional to the number of IDT fingers. In contrast, energy lost from the ends of the IDT in the longitudinal direction (i.e. the direction normal to the IDT fingers) is roughly constant, independent of the number of IDT fingers. As the number of IDT fingers and the peak energy stored in an XBAR is reduced, the acoustic energy lost in the longitudinal direction becomes an ever-increasing fraction of the peak energy stored.

[0044] FIG. **3** is a graph of the normalized Q factor of a representative XBAR as a function of the number of fingers in the IDT of the XBAR. The “normalized Q-factor” is the Q-factor of the XBAR with a finite number of IDT fingers divided by the Q-factor of a hypothetical XBAR with the same structure and an infinite number of IDT fingers. In FIG. **3**, normalized Q-factor is quantified as a percentage of the Q-factor of the XBAR with an infinite number of IDT fingers. Specifically, the solid curve **310** is a plot of the normalized Q-factor at the resonance frequency and the dashed curve **320** is a plot of the normalized Q-factor at the anti-resonance frequency. The data in FIG. **3** is derived from simulations using a finite element method.

[0045] FIG. **3** shows that the normalized Q-factor of an XBAR with a finite number of IDT fingers is less than 100%, which is to say the Q-factor of an XBAR with a finite number of IDT fingers is less than the Q-factor of a similar XBAR with an infinite number of IDT fingers. Although not shown in FIG. **3**, the normalized Q-factor of an XBAR may asymptotically approach 100% for a very large number of IDT fingers. As anticipated, the normalized Q-factor depends on the number of IDT fingers. In particular, the normalized Q-factor decreases precipitously for XBARs with less than about 20 IDT fingers due to the increasing significance of acoustic energy lost in the longitudinal direction.

[0046] FIG. **4** is a plan view of an exemplary conductor pattern **400** that reduces the acoustic energy leakage in the longitudinal direction at the ends of an XBAR. The conductor pattern **400** includes an IDT **430** and four reflector elements **462**, **464**, **466**, **468**. The IDT **430** includes a first busbar **432**, a second busbar **434**, and a plurality of n interleaved IDT fingers extending alternately from the first and second busbars. In this example, n , the number of IDT fingers, is equal to 24. In other XBARs, n may be in a range from 20 to 100 or more IDT fingers. IDT finger **436** is the

1.sup.st finger and IDT finger **438** is the n'th finger. Numbering the IDT fingers from left to right (as shown in FIG. **4**) is arbitrary and the designations of the 1.sup.st and n'th fingers could be reversed.

[0047] As shown in FIG. **4**, the odd numbered IDT fingers extend from the first busbar **432** and the even numbered IDT fingers extend from the second busbar **434**. The IDT **430** has an even number of IDT fingers such that the 1.sup.st and n'th IDT fingers **436**, **438** extend from different busbars. In some cases, an IDT may have an odd number of IDT fingers such that the 1.sup.st and n'th IDT fingers and all of the reflector elements extend from the same busbar.

[0048] A total of four reflector elements are provided outside of periphery of the IDT **430**. A first reflector element **462** is proximate and parallel to 1st IDT finger **436** at the left end of the IDT **430**. A second reflector element **466** is proximate and parallel to n'th IDT finger **438** at the right end of the IDT **430**. An optional third reflector element **464** is parallel to the first reflector element **462**. An optional fourth reflector element **468** is parallel to the second reflector element **466**.

[0049] First and third reflector elements **462**, **464** extend from the first busbar **432** and thus are at the same electrical potential as the 1st IDT finger **436**. Similarly, second and fourth reflector elements **466** and **468** extend from the second busbar **430** and thus are at the same electrical potential as the n'th IDT finger **438**.

[0050] The reflector elements **462**, **464**, **466**, **468** are configured to confine acoustic energy to the area of the IDT **430** and thus reduce acoustic energy losses in the longitudinal direction. To this end, the pitch p_r between adjacent reflector elements and between reflector elements **462** and **466** and the adjacent first and n'th IDT fingers, respectively, is typically greater than the pitch p of the IDT fingers. The width or mark m_r of the reflector elements **462**, **464**, **466**, **468** is not necessarily equal to the mark m of the IDT fingers. As will be described subsequently, the mark m_r of the reflector elements may be selected to optimize Q-factor at a specific frequency or range of frequencies.

[0051] FIG. **5** is a plan view of another conductor pattern **500** that reduces the acoustic energy leakage in the longitudinal direction at the ends of an XBAR. The conductor pattern **500** includes an IDT **530** and four reflector elements **562**, **564**, **566**, **568**. The IDT **530** includes a first busbar **532**, a second busbar **534**, and a plurality of interleaved IDT fingers extending alternately from the first and second busbars as previously described. IDT fingers **536** and **538** are the 1.sup.st and n'th IDT fingers at the left and right (as shown in FIG. **5**) ends of the IDT **530**.

[0052] A total of four reflector elements are provided outside of periphery of the IDT **530**. First and third reflector elements **562** and **564** are proximate and parallel to 1st IDT finger **536** at the left end of the IDT **530**. First and third reflector elements **562**, **564** are connected to each other but are not connected to either busbar **532**, **534**. First and third reflector elements **562**, **564** are capacitively coupled to 1st IDT finger **536** and thus are at substantially the same electrical potential as the 1st IDT finger **536**. The reflector elements are considered to be at substantially the same potential if, when an RF signal is applied between the busbars **532**, **534**, the potential between the reflector elements and the 1.sup.st IDT finger is small compared to the potential between adjacent IDT fingers.

[0053] Similarly, second and fourth reflector elements **566** and **568** are proximate and parallel to n'th IDT finger **538** at the right end of the IDT **530**. Second and fourth reflector elements **566**, **568** are connected to each other and not connected to each other or either busbar **532**, **534**. Second and fourth reflector elements **566**, **568** are capacitively coupled to each other and to n'th IDT finger **538** and thus are at nearly the same electrical potential as the n'th IDT finger **538**.

[0054] The reflector elements **562**, **564**, **566**, **568** are configured to confine acoustic energy to the area of the IDT **530** and thus reduce acoustic energy lost in the longitudinal direction. To this end, the pitch p_r between adjacent reflector elements and between reflector elements **562** and **566** and the adjacent terminal IDT fingers is typically greater than the pitch p of the IDT fingers. The width or mark m_r of the reflector elements **562**, **564**, **566**, **568** is not necessarily equal to the mark m of

the IDT fingers. The mark m_r of the reflector elements may be selected to optimize Q-factor for a particular frequency or range of frequencies.

[0055] FIG. 6 is a graph of the normalized Q-factor as a function of number of IDT fingers for another XBAR with and without reflector elements similar to the reflector elements shown in FIG. 4. Specifically, the solid curve 610 is a plot of the normalized Q-factor of an XBAR without reflector elements at its resonance frequency. The dashed curve 620 is a plot of the normalized Q-factor at the resonance frequency for a similar XBAR with two reflector elements at each side of the IDT. In both cases, the piezoelectric plate is lithium niobate 400 nm thick, the IDT fingers are aluminum 500 nm thick, IDT pitch $p=4$ microns, and IDT finger mark $m=1$ micron. For the XBAR with reflector elements, $p_r=4.2$ microns and $m_r=0.735$ microns. With reflector elements, an XBAR with as few as 10 fingers can have a normalized Q-factor up to 80%.

[0056] FIG. 7 is a graph of the normalized Q-factor as a function of number of IDT fingers for another XBAR with and without reflector elements similar to the reflector elements shown in FIG. 4. Specifically, the solid curve 710 is a plot of the normalized Q-factor of an XBAR without reflector elements at its anti-resonance frequency. The dashed curve 720 is a plot of the normalized Q-factor at the anti-resonance frequency for a similar XBAR with two reflector elements at each side of the IDT. In both cases, the piezoelectric plate is lithium niobate 400 nm thick, the IDT fingers are aluminum 500 nm thick, IDT pitch $p=4$ microns, and IDT finger mark $m=1$ micron. For the XBAR with reflector elements, $p_r=8$ microns and $m_r=0.80$ microns. With reflector elements, an XBAR with as few as 14 fingers can have a normalized Q-factor up to 80%.

[0057] FIG. 8 shows a chart 800 illustrating the relationship between the pitch p_r and mark m_r of reflector elements of an exemplary XBAR device at a fixed frequency of 5150 MHz. The exemplary XBAR device has a lithium niobate piezoelectric plate with a thickness of 400 nm and aluminum IDT and reflector elements with a thickness of 500 nm. The pitch and mark of the IDT fingers are 4 microns and 1 micron, respectively. There are two reflector elements at each end of the IDT. The lighter shaded area 810A, 810B, 810C, 810D identify combinations of p_r and m_r where the normalized Q-factor is greater than or equal to 85%. The darker shaded area 820A, 820B, 820C, 820D identify combinations of p_r and m_r where the normalized Q-factor is greater than or equal to 90%. For comparison, the normalized Q-factor of this XBAR without reflector elements is 74% at 5150 MHz. Although not identified in FIG. 8 there are combinations of p_r and m_r where the normalized Q-factor is less than 75%, indicating improperly configured reflector elements can degrade XBAR Q-factor.

[0058] There are multiple combinations of p_r and m_r that raise the normalized Q-factor to 85% or 90%. To achieve a normalized Q-factor of greater than or equal to 90%, p_r must be greater than or equal to 1.2 times the pitch p of the IDT fingers. For $p_r=6$ microns ($1.5p$), there are at least four values of m_r that raise the normalized Q-factor to more than 90%.

[0059] FIG. 9 shows a chart 900 illustrating the relationship between the mark m_r of reflector elements and frequency for an exemplary XBAR device with two reflector elements at each side of an IDT and $p_r=5.2$ microns. As in the previous examples, the exemplary XBAR device has a lithium niobate piezoelectric plate with a thickness of 400 nm and aluminum IDT and reflector elements with a thickness of 500 nm. The pitch and mark of the IDT fingers are 4 microns and 1 micron, respectively. Lighter shaded areas such as area 910 identify combinations of frequency and m_r where the normalized Q-factor is greater than or equal to 85%. Darker shaded areas such as area 920 identify combinations of frequency and m_r where the normalized Q-factor is greater than or equal to 90%. For comparison, the normalized Q-factor of this XBAR without reflector elements is 74% at 5150 MHz.

[0060] The chart 900 illustrates that, for a particular reflector element pitch p_r , reflector element mark m_r must be selected with consideration of the frequency where the Q-factor of an XBAR needs improvement. For example, selecting $m_r=0.95$ microns provides a normalized Q-factor greater than 90% over a frequency range from about 4980 MHz to greater than 5200 MHz.

Selecting $mr=1.7$ microns provides a normalized Q-factor greater than 90% over a frequency range of less than 4700 MHz to about 4950 MHz. However, selecting $mr=1.7$ microns may actually lower the Q-factor at 5200 MHz compared to an XBAR with no reflector elements.

[0061] FIG. **10** shows a chart **1100** illustrating the relationship between the mark mr of reflector elements and frequency for an exemplary XBAR device with one reflector element at each side of an IDT and $pr=5.2$ microns. The exemplary XBAR device is the same as the previous examples. As in FIG. **9**, lighter shaded areas such as area **1010** identify combinations of frequency and mr where the normalized Q-factor is greater than or equal to 85%. Darker shaded areas such as area **1020** identify combinations of frequency and mr where the normalized Q-factor is greater than or equal to 90%.

[0062] Comparison of FIG. **9** and FIG. **10** shows that a only one reflector element is generally less effective than two reflector elements for at improving normalized Q-factor. However, in some applications one reflector element at each end of an IDT may be sufficient. In this example, one reflector element (at each end of the IDT) with $mr=0.75$ microns provides a substantial improvement in normalized Q-factor for a frequency range of about 4770 MHz to 4970 MHz.

[0063] FIG. **11** shows a chart **1100** illustrating the relationship between the mark mr of reflector elements and frequency for an exemplary XBAR device with five reflector elements at each end of an IDT and $pr=5.2$ microns. The exemplary XBAR device is the same as the previous examples. As in FIG. **9**, lighter shaded areas such as area **1110** identify combinations of frequency and mr where the normalized Q-factor is greater than or equal to 85%. Darker shaded areas such as area **1120** identify combinations of frequency and mr where the normalized Q-factor is greater than or equal to 90%.

[0064] Comparison of FIG. **9** and FIG. **11** shows that five reflector elements do not provide any significant improvement over two reflector elements.

[0065] FIG. **12** is a chart of the performance of an exemplary XBAR bandpass filter with and without reflector elements. Specifically, the solid line **1210** is a plot of the magnitude of $S_{sub.21}$ (the input-output transfer function) of a Band n77 filter with two reflector elements at each end of the XBARs in the filter. The reflector elements of shunt resonators were optimized for a frequency of 3.35 GHz and the reflector elements of series resonators were optimized for a frequency of 4.2 GHz. These frequencies are at or near the edges of the n77 band where it is typically most difficult to achieve requirements on minimum $S_{sub.21}$. The dashed line **1220** is a plot of the magnitude of $S_{sub.21}$ for the same filter without reflector elements on the XBARs. All of the data was developed by simulations of the filters using a finite element method.

[0066] The inclusion of reflector elements improves $S_{sub.21}$ by 0.2 db at 3.35 GHz and by 0.4 dB at 4.2 GHz. Note, however, that the inclusion of the reflector elements reduces $S_{sub.21}$ by as much as 0.25 dB at other frequencies, illustrating a tradeoff to be made during the design of an XBAR filter. In the exemplary bandpass filter of FIG. **12**, the reflector elements for all of the shunt resonators were selected for maximum Q-factor at the same frequency (3.35 GHz) and the reflector elements for all of the series resonators were selected for maximum Q-factor at the same frequency (4.2 GHz). Further improvements in the filter transfer function are likely if the reflector elements for each resonator are optimized independently.

[0067] FIG. **13** is a plan view of another exemplary conductor pattern **1300** that reduces the acoustic energy leakage in the longitudinal direction at the ends of an XBAR. The conductor pattern **1300** includes an IDT **1330** and first, second, third, and fourth reflector elements **1362**, **1364**, **1366**, **1368**. The IDT **1330** includes a first busbar **1332**, a second busbar **1334**, and a plurality of interleaved IDT fingers extending alternately from the first and second busbars. In this example, number of IDT fingers, is equal to 24. In other XBARs, the number of fingers may be in a range from 20 to 100 or more IDT fingers. Overlapping portions of the interleaved IDT fingers and the first, second, third, and fourth reflector elements **1362**, **1364**, **1366**, **1368** are on the diaphragm of an XBAR. IDT finger **1336** is the first finger and IDT finger **1338** is the last finger. The

designations of the first and last IDT fingers is arbitrary and the designations of the first and last fingers could be reversed.

[0068] A total of four reflector elements are provided outside of periphery of the IDT **1330**. The first and second reflector elements **1362**, **1364** are proximate and parallel to the first IDT finger **1336** at the left end (as shown in the figure) of the IDT **1330**. The third and fourth reflector elements **1366**, **1368** are proximate and parallel to the last IDT finger **1338** at the right end (as shown) of the IDT **1330**.

[0069] First and second reflector elements **1362**, **1364** extend from a different busbar than the first IDT finger. In this example, the first and second reflector elements **1362**, **1364** extend from the second busbar **1334** and the first IDT finger **1336** extends from the first busbar **1332**. Thus, the electrical potential on the first and second reflector elements **1362**, **1364** is opposite of the potential on the first IDT finger **1336**. Similarly, third and fourth reflector elements **1366**, **1368** extend a different busbar than the last IDT finger **1338**. Thus, the electrical potential on the third and fourth reflector elements **1366**, **1368** is opposite of the potential on the last IDT finger **1338**.

[0070] The reflector elements **1362**, **1364**, **1366**, **1368** are configured to confine acoustic energy to the area of the IDT **1330** and thus reduce acoustic energy losses in the longitudinal direction. To this end, the center-to-center distance $pr1$ between adjacent reflector elements (i.e., between the first and second reflector elements and between the third and four reflector elements) is typically greater than or equal to 1.1 times the pitch p of the IDT fingers and less than or equal to 1.5 times p .

[0071] The center-to-center distance $pr2$ between reflector elements and the adjacent IDT fingers (i.e. between the first reflector element and the first IDT finger, and between the third reflector element and the last IDT finger) is typically greater than p and less than or equal to $pr1$. In some cases, $pr2$ may be the average of p and $pr1$.

[0072] The width or mark mr of the reflector elements **1362**, **1364**, **1366**, **1368** is not necessarily equal to the mark m of the IDT fingers. As was described previously, the mark mr of the reflector elements may be selected to optimize Q-factor at a specific frequency or range of frequencies.

[0073] FIG. **14A** is graph **1400** of the magnitude of S_{21} (the input/output transfer function) as a function of frequency for a first pair of XBAR filters with and without reflector fingers. The filters are designed to pass 5G NR band n79. The band n79 frequency range is identified.

[0074] The solid curve **1410** is the magnitude of S_{21} for a filter with reflector fingers on all series resonators. For each series resonator, $pr1=1.3p$ and $pr2=1.15p$. mr was independently optimized for each resonator to minimize loss at the upper band edge (5.0 GHz). The dashed curve **1420** is the magnitude of S_{21} for the same filter without reflector fingers on any resonator. The presence of reflector fingers on the series resonators (solid curve **1410**) reduces losses (i.e. increases S_{21}) at the upper band edge by about 0.4 dB compared to the filter without reflector fingers (dashed curve **1420**).

[0075] FIG. **14B** is graph **1450** of the magnitude of S_{21} (the input/output transfer function) as a function of frequency for a second pair of XBAR filters with and without reflector fingers. The filters are designed, independently from the design of the first pair of filters, to pass 5G NR band n79. The band n79 frequency range is identified.

[0076] The solid curve **1460** is the magnitude of S_{21} for a filter with reflector fingers on all series resonators. For each series resonator, $pr1=1.3p$ and $pr2=1.15p$. mr was independently optimized for each resonator to minimize loss at the upper band edge (5.0 GHz). The dashed curve **1470** is the magnitude of S_{21} for the same filter without reflector fingers on any resonator. The presence of reflector fingers on the series resonators (solid curve **1460**) reduces losses (i.e. increases S_{21}) at the upper band edge by about 0.3 dB compared to the filter without reflector fingers (dashed curve **1470**).

[0077] FIG. **14A** and FIG. **14B** showed performance data for two exemplary filters with two reflector elements at each end of the IDT of each series resonator. In both examples, the center-to-

center distance $pr1$ between the pairs of reflector elements was $1.3p$ and the center-to-center distance $pr2$ between the end finger of the IDT and the adjacent reflector element is $0.5(p+pr1)$. Other filters may use only one reflector element at the ends of some or all IDTs. Other filters may include reflector elements fewer than all series resonators and/or include reflector elements on some or all shunt resonators. When reflector elements are used on both series and shunt resonators, the number of elements (i.e. 1 or 2) and the distances $pr1$, $pr2$ may be different for shunt and series resonator. For example, a filter may use two reflector elements at the ends of series resonators with $pr1=pr2=1.3p$ and one reflector elements at the ends of shunt resonators with $pr2=1.1p$. In all cases, the width or mark of the reflector elements may be optimized independently for each resonator.

Description of Methods

[0078] FIG. 15 is a simplified flow chart summarizing a process 1500 for fabricating a filter device incorporating XBARS. Specifically, the process 1500 is for fabricating a filter device including multiple XBARS, some of which may include a frequency setting dielectric layer. The process 1500 starts at 1505 with a device substrate and a thin plate of piezoelectric material disposed on a sacrificial substrate. The process 1500 ends at 1595 with a completed filter device. The flow chart of FIG. 15 includes only major process steps. Various conventional process steps (e.g. surface preparation, cleaning, inspection, baking, annealing, monitoring, testing, etc.) may be performed before, between, after, and during the steps shown in FIG. 15.

[0079] While FIG. 15 generally describes a process for fabricating a single filter device, multiple filter devices may be fabricated simultaneously on a common wafer (consisting of a piezoelectric plate bonded to a substrate). In this case, each step of the process 1500 may be performed concurrently on all of the filter devices on the wafer.

[0080] The flow chart of FIG. 15 captures three variations of the process 1500 for making an XBAR which differ in when and how cavities are formed in the device substrate. The cavities may be formed at steps 1510A, 1510B, or 1510C. Only one of these steps is performed in each of the three variations of the process 1500.

[0081] The piezoelectric plate may be, for example, lithium niobate or lithium tantalate, either of which may be Z-cut, rotated Z-cut, or rotated YX-cut. The piezoelectric plate may be some other material and/or some other cut. The device substrate may preferably be silicon. The device substrate may be some other material that allows formation of deep cavities by etching or other processing.

[0082] In one variation of the process 1500, one or more cavities are formed in the device substrate at 1510A, before the piezoelectric plate is bonded to the substrate at 1515. A separate cavity may be formed for each resonator in a filter device. The one or more cavities may be formed using conventional photolithographic and etching techniques. Typically, the cavities formed at 1510A will not penetrate through the device substrate.

[0083] At 1515, the piezoelectric plate is bonded to the device substrate. The piezoelectric plate and the device substrate may be bonded by a wafer bonding process. Typically, the mating surfaces of the device substrate and the piezoelectric plate are highly polished. One or more layers of intermediate materials, such as an oxide or metal, may be formed or deposited on the mating surface of one or both of the piezoelectric plate and the device substrate. One or both mating surfaces may be activated using, for example, a plasma process. The mating surfaces may then be pressed together with considerable force to establish molecular bonds between the piezoelectric plate and the device substrate or intermediate material layers.

[0084] At 1520, the sacrificial substrate may be removed. For example, the piezoelectric plate and the sacrificial substrate may be a wafer of piezoelectric material that has been ion implanted to create defects in the crystal structure along a plane that defines a boundary between what will become the piezoelectric plate and the sacrificial substrate. At 1520, the wafer may be split along the defect plane, for example by thermal shock, detaching the sacrificial substrate and leaving the piezoelectric plate bonded to the device substrate. The exposed surface of the piezoelectric plate

may be polished or processed in some manner after the sacrificial substrate is detached.

[0085] Thin plates of single-crystal piezoelectric materials laminated to a non-piezoelectric substrate are commercially available. At the time of this application, both lithium niobate and lithium tantalate plates are available bonded to various substrates including silicon, quartz, and fused silica. Thin plates of other piezoelectric materials may be available now or in the future. The thickness of the piezoelectric plate may be between 300 nm and 1000 nm. When the substrate is silicon, a layer of SiO₂ may be disposed between the piezoelectric plate and the substrate. When a commercially available piezoelectric plate/device substrate laminate is used, steps **1510A**, **1515**, and **1520** of the process **1500** are not performed.

[0086] A first conductor pattern, including IDTs and reflector elements of each XBAR, is formed at **1545** by depositing and patterning one or more conductor layers on the front side of the piezoelectric plate. The conductor layer may be, for example, aluminum, an aluminum alloy, copper, a copper alloy, or some other conductive metal. Optionally, one or more layers of other materials may be disposed below (i.e. between the conductor layer and the piezoelectric plate) and/or on top of the conductor layer. For example, a thin film of titanium, chrome, or other metal may be used to improve the adhesion between the conductor layer and the piezoelectric plate. A second conductor pattern of gold, aluminum, copper or other higher conductivity metal may be formed over portions of the first conductor pattern (for example the IDT bus bars and interconnections between the IDTs).

[0087] Each conductor pattern may be formed at **1545** by depositing the conductor layer and, optionally, one or more other metal layers in sequence over the surface of the piezoelectric plate. The excess metal may then be removed by etching through patterned photoresist. The conductor layer can be etched, for example, by plasma etching, reactive ion etching, wet chemical etching, or other etching techniques.

[0088] Alternatively, each conductor pattern may be formed at **1545** using a lift-off process. Photoresist may be deposited over the piezoelectric plate. and patterned to define the conductor pattern. The conductor layer and, optionally, one or more other layers may be deposited in sequence over the surface of the piezoelectric plate. The photoresist may then be removed, which removes the excess material, leaving the conductor pattern.

[0089] At **1550**, one or more frequency setting dielectric layer(s) may be formed by depositing one or more layers of dielectric material on the front side of the piezoelectric plate. For example, a dielectric layer may be formed over the shunt resonators to lower the frequencies of the shunt resonators relative to the frequencies of the series resonators. The one or more dielectric layers may be deposited using a conventional deposition technique such as physical vapor deposition, atomic layer deposition, chemical vapor deposition, or some other method. One or more lithography processes (using photomasks) may be used to limit the deposition of the dielectric layers to selected areas of the piezoelectric plate. For example, a mask may be used to limit a dielectric layer to cover only the shunt resonators.

[0090] At **1555**, a passivation/tuning dielectric layer is deposited over the piezoelectric plate and conductor patterns. The passivation/tuning dielectric layer may cover the entire surface of the filter except for pads for electrical connections to circuitry external to the filter. In some instantiations of the process **1500**, the passivation/tuning dielectric layer may be formed after the cavities in the device substrate are etched at either **1510B** or **1510C**.

[0091] In a second variation of the process **1500**, one or more cavities are formed in the back side of the device substrate at **1510B**. A separate cavity may be formed for each resonator in a filter device. The one or more cavities may be formed using an anisotropic or orientation-dependent dry or wet etch to open holes through the back side of the device substrate to the piezoelectric plate. In this case, the resulting resonator devices will have a cross-section as shown in FIG. 1.

[0092] In a third variation of the process **1500**, one or more cavities in the form of recesses in the device substrate may be formed at **1510C** by etching the substrate using an etchant introduced

through openings in the piezoelectric plate. A separate cavity may be formed for each resonator in a filter device. The one or more cavities formed at **1510C** will not penetrate through the device substrate.

[0093] Ideally, after the cavities are formed at **1510B** or **1510C**, most or all of the filter devices on a wafer will meet a set of performance requirements. However, normal process tolerances will result in variations in parameters such as the thicknesses of dielectric layer formed at **1550** and **1555**, variations in the thickness and line widths of conductors and IDT fingers formed at **1545**, and variations in the thickness of the piezoelectric plate. These variations contribute to deviations of the filter device performance from the set of performance requirements.

[0094] To improve the yield of filter devices meeting the performance requirements, frequency tuning may be performed by selectively adjusting the thickness of the passivation/tuning layer deposited over the resonators at **1555**. The frequency of a filter device passband can be lowered by adding material to the passivation/tuning layer, and the frequency of the filter device passband can be increased by removing material to the passivation/tuning layer. Typically, the process **1500** is biased to produce filter devices with passbands that are initially lower than a required frequency range but can be tuned to the desired frequency range by removing material from the surface of the passivation/tuning layer.

[0095] At **1560**, a probe card or other means may be used to make electrical connections with the filter to allow radio frequency (RF) tests and measurements of filter characteristics such as input-output transfer function. Typically, RF measurements are made on all, or a large portion, of the filter devices fabricated simultaneously on a common piezoelectric plate and substrate.

[0096] At **1565**, global frequency tuning may be performed by removing material from the surface of the passivation/tuning layer using a selective material removal tool such as, for example, a scanning ion mill as previously described. “Global” tuning is performed with a spatial resolution equal to or larger than an individual filter device. The objective of global tuning is to move the passband of each filter device towards a desired frequency range. The test results from **1560** may be processed to generate a global contour map indicating the amount of material to be removed as a function of two-dimensional position on the wafer. The material is then removed in accordance with the contour map using the selective material removal tool.

[0097] At **1570**, local frequency tuning may be performed in addition to, or instead of, the global frequency tuning performed at **1565**. “Local” frequency tuning is performed with a spatial resolution smaller than an individual filter device. The test results from **1560** may be processed to generate a map indicating the amount of material to be removed at each filter device. Local frequency tuning may require the use of a mask to restrict the size of the areas from which material is removed. For example, a first mask may be used to restrict tuning to only shunt resonators, and a second mask may be subsequently used to restrict tuning to only series resonators (or vice versa). This would allow independent tuning of the lower band edge (by tuning shunt resonators) and upper band edge (by tuning series resonators) of the filter devices.

[0098] After frequency tuning at **1565** and/or **1570**, the filter device is completed at **1575**. Actions that may occur at **1575** include forming bonding pads or solder bumps or other means for making connection between the device and external circuitry (if such pads were not formed at **1545**); excising individual filter devices from a wafer containing multiple filter devices; other packaging steps; and additional testing. After each filter device is completed, the process ends at **1595**.

Closing Comments

[0099] Throughout this description, the embodiments and examples shown should be considered as exemplars, rather than limitations on the apparatus and procedures disclosed or claimed. Although many of the examples presented herein involve specific combinations of method acts or system elements, it should be understood that those acts and those elements may be combined in other ways to accomplish the same objectives. With regard to flowcharts, additional and fewer steps may be taken, and the steps as shown may be combined or further refined to achieve the methods

described herein. Acts, elements and features discussed only in connection with one embodiment are not intended to be excluded from a similar role in other embodiments.

[0100] As used herein, “plurality” means two or more. As used herein, a “set” of items may include one or more of such items. As used herein, whether in the written description or the claims, the terms “comprising”, “including”, “carrying”, “having”, “containing”, “involving”, and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of”, respectively, are closed or semi-closed transitional phrases with respect to claims. Use of ordinal terms such as “first”, “second”, “third”, etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements. As used herein, “and/or” means that the listed items are alternatives, but the alternatives also include any combination of the listed items.

Claims

1. An acoustic resonator comprising: a piezoelectric layer; a conductor pattern on a surface of the piezoelectric layer, the conductor pattern comprising: an interdigital transducer (IDT) comprising a first busbar, a second busbar, and interleaved fingers that each extend from one of the first and second busbars, the interleaved fingers comprising a first finger and a last finger at opposing ends of the IDT; a first reflector element proximate to the first finger; and a second reflector element proximate to the last finger, wherein a distance between the first reflector element and the first finger is greater than a pitch p of the IDT, the pitch p being a center-to-center spacing between at least two adjacent interleaved fingers of the interleaved fingers of the IDT, wherein a distance between the second reflector element and the last finger is greater than the pitch p of the IDT.
2. The acoustic resonator of claim 1, wherein one or more of the distance between the first reflector element and the first finger and the distance between the second reflector element and the last finger is greater than or equal to $1.2p$ and less than or equal to $1.5p$.
3. The acoustic resonator of claim 1, wherein a width m_r of each of the first and second reflector elements is configured to improve a Q-factor of the acoustic resonator at a predetermined frequency.
4. The acoustic resonator of claim 1, wherein the first reflector element is connected to a same one of the first busbar and the second busbar as the first finger, and the second reflector element is connected to a same one of the first busbar and the second busbar as the last finger.
5. The acoustic resonator of claim 1, wherein the first reflector element is connected to an opposite one of the first busbar and the second busbar as the first finger, and the second reflector element is connected to an opposite one of the first busbar and the second busbar as the last finger.
6. The acoustic resonator of claim 1, wherein either the first reflector extends from one of the first busbar or the second busbar, and the second reflector extends from one of the first busbar or the second busbar.
7. The acoustic resonator of claim 1, further comprising a third reflector element disposed such that the first reflector element is between the third reflector element and the first finger, and wherein a distance between the first reflector element and the third reflector element is greater than the pitch p of the IDT.
8. The acoustic resonator of claim 7, wherein the distance between the first reflector element and the third reflector element is greater than equal to $1.2p$ and less than or equal to $1.5p$.
9. The acoustic resonator of claim 1, further comprising a fourth reflector element disposed such that the second reflector element is between the fourth reflector element and the last finger, and wherein a distance between the second reflector element and the fourth reflector element is greater

than the pitch p of the IDT.

10. The acoustic resonator of claim 9, wherein the distance between the second reflector element and the fourth reflector element is greater than or equal to $1.2p$ and less than or equal to $1.5p$.

11. The acoustic resonator of claim 7, wherein the distance between the first reflector element and the first finger is substantially equal to a sum, divided by two, of the pitch p and the distance between the first reflector element and the third reflector element.

12. A filter device comprising: a plurality of bulk acoustic resonators, wherein at least one bulk acoustic resonator of the plurality of bulk acoustic resonators comprises: a piezoelectric layer; a conductor pattern on a surface of the piezoelectric layer, the conductor pattern comprising: an interdigital transducer (IDT) comprising a first busbar, a second busbar, and interleaved fingers that each extend from one of the first and second busbars, the interleaved fingers comprising a first finger and a last finger at opposing ends of the IDT; a first reflector element proximate to the first finger; and a second reflector element proximate to the last finger, wherein a distance between the first reflector element and the first finger is greater than a pitch p of the IDT, the pitch p being a center-to-center spacing between at least two adjacent interleaved fingers of the interleaved fingers of the IDT, wherein a distance between the second reflector element and the last finger is greater than the pitch p of the IDT, wherein the first reflector extends from one of the first busbar or the second busbar, and the second reflector extends from one of the first busbar or the second busbar.

13. The filter device of claim 12, wherein one or more of the distance between the first reflector element and the first finger and the distance between the second reflector element and the last finger is greater than or equal to $1.2p$ and less than or equal to $1.5p$.

14. The filter device of claim 12, wherein the at least one bulk acoustic resonator comprises a shunt bulk acoustic resonator and a series bulk acoustic resonator, and wherein the distance between the first reflector element and the first finger for the series resonator is different than the distance between the first reflector element and the first finger for the shunt bulk acoustic resonator.

15. The filter device of claim 12, wherein: the at least one bulk acoustic resonator comprises a shunt bulk acoustic resonator and a series bulk acoustic resonator, the shunt bulk acoustic resonator and the series bulk acoustic resonator each comprise a third reflector element disposed such that the first reflector element is between the third reflector element and the first finger, and the distance between the first and third reflector elements is different between the series bulk acoustic resonator and the shunt bulk acoustic resonator.

16. The filter device of claim 12, wherein the at least one bulk acoustic resonator comprises a shunt bulk acoustic resonator and a series bulk acoustic resonator, and wherein the distance between the first reflector element and the first finger for the series resonator is more than the distance between the first reflector element and the first finger for the shunt bulk acoustic resonator.

17. The filter device of claim 12, wherein the first reflector element is connected to a same one of the first busbar and the second busbar as the first finger, and the second reflector element is connected to a same one of the first busbar and the second busbar as the last finger.

18. The filter device of claim 12, wherein the first reflector element is connected to an opposite one of the first busbar and the second busbar as the first finger, and the second reflector element is connected to an opposite one of the first busbar and the second busbar as the last finger.

19. The filter device of claim 12, further comprising a third reflector element disposed such that the first reflector element is between the third reflector element and the first finger, and wherein a distance between the first reflector element and the third reflector element is greater than the pitch p of the IDT.

20. The filter device of claim 19, wherein the distance between the first reflector element and the third reflector element is greater than or equal to $1.2p$ and less than or equal to $1.5p$.
